



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12 2008 HALF YEARLY EXAMINATION

EXTENSION II MATHEMATICS

STUDENT NUMBER: _____

ASSESSMENT TASK 3

Time Allowed – 3 hours (*Plus 5 minutes Reading Time*)

Total marks – 120

WEIGHTING 30% towards final result

Outcomes referred to: E1, E2, E6, E8 and E9 (refer booklet)

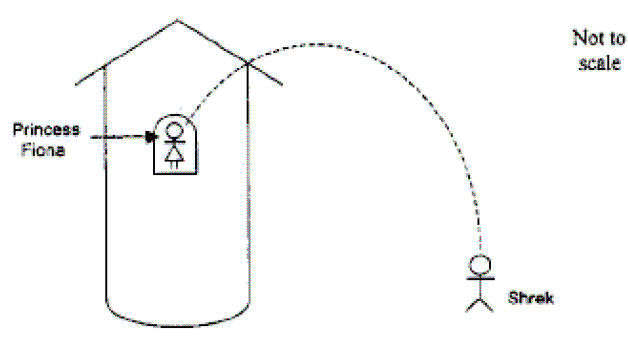
INSTRUCTIONS:

1. Attempt **ALL** questions.
2. Show all necessary working.
3. Put your Student Number **ONLY** on each writing sheet.
4. Begin **each** question on a **new page**.
5. Marks are shown next to each question.
6. Additional writing sheets may be obtained from the supervisor upon request.
7. Only approved Board of Studies calculators are permitted.
8. A Standard Integrals sheet is provided on a separate sheet.

Question 1:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) (i) A particle is moving along the x -axis so that at any time $t \geq 0$ its velocity is v and its acceleration is $\frac{dv}{dt}$. Prove that $\frac{dv}{dt} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$ 2
- (ii) A particle is moving along the x -axis in simple harmonic motion with its velocity given by the equation $v^2 = 4x - 2x^2$. Find
- (α) Its amplitude. 2
- (β) Its period. 2

(b)



Princess Fiona is locked up in a tower, 80m above the ground.

To gain the attention of Shrek, Princess Fiona throws a lenticil at an angle of elevation of θ and an initial velocity of 50ms^{-1} .

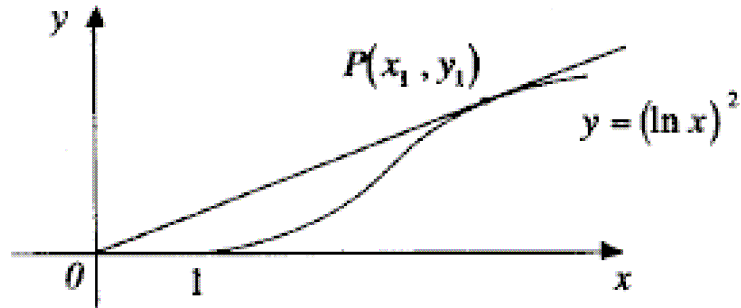
- (i) Derive the equations for the horizontal and vertical displacements of the lenticil t seconds after it is thrown. (Use $g = 10\text{ms}^{-2}$.) 4
- (ii) Shrek is 300m from the base of the tower when he is hit by the lenticil. Find the values of the initial angle of projection θ , correct to the nearest degree, if Shrek is 2m tall. 5

Question 2:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) It is given that $(2 + \cos x)(2 - \cos y) = 3$, where $0 < x < \pi$ and $0 < y < \pi$.
- (i) Show that $\cos y = \frac{1 + 2 \cos x}{2 + \cos x}$ and $\sin y = \frac{\sqrt{3} \sin x}{2 + \cos x}$. **2**
- (ii) Hence show that $\frac{dy}{dx} = \frac{\sqrt{3}}{2 + \cos x}$ **3**
- (b) The function $f(x)$ is given by $f(x) = a + \frac{b \sin x}{x}$, $x \neq 0$ and $f(0) = 0$, where a and b are non-zero real numbers.
- (i) Show that $f(x)$ is an even function. **1**
- (ii) Find the general solution of the equation $f(x) = a$. **2**
- (iii) If $\lim_{x \rightarrow \infty} f(x) = 1$ and $f(x)$ is continuous at $x = 0$, find the values of a and b . **2**
- (c) Find $\int \frac{1}{1-x^2} dx$. **2**
- (d) Use the substitution $u = e^x - 1$ to find $\int \frac{e^{2x}}{(e^x - 1)^2} dx$. **3**

Question 3:*Use a SEPARATE Writing Booklet.***Total Marks 15**

(a)



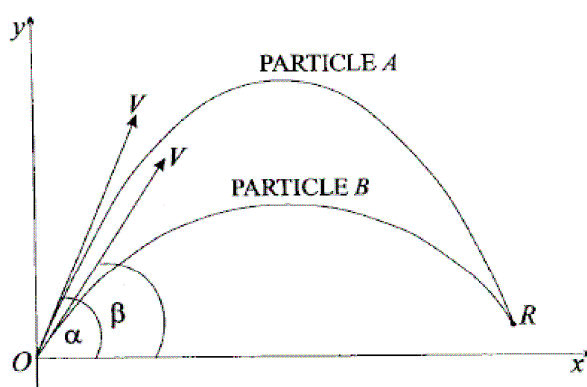
The diagram shows the graph of the function $f(x) = (\ln x)^2$.

$P(x_1, y_1)$ is a point on the curve such that the tangent to the curve at P passes through the origin O.

- (i) By considering the gradient of the line OP in two different ways, show that P is the point $(e^2, 4)$. 2
- (ii) Find the set of values of the real number k such that the equation $f(x) = kx$ has two distinct real roots. 1
- (iii) Use integration by parts to show that
- $$\int (\ln x)^2 dx = x(\ln x)^2 - 2x \ln x + 2x + c.$$
- Hence find the exact area of the region bounded by the curve $y = f(x)$, the x axis and the line OP. 4
- (iv) Find the equation of the inverse function $f^{-1}(x)$. 1
- (v) Find the equation of the tangent to the curve $y = f^{-1}(x)$ that passes through the origin. 1

Question 3 continued next page

(b) (continued from previous page)



The diagram above shows two particles A and B projected from the origin. Particle A is projected with an initial velocity of $V \text{ ms}^{-1}$ at an angle α . Particle B is projected T seconds later with the same initial velocity $V \text{ ms}^{-1}$ but at an angle of β . The particles collide at a point R.

(i) You may assume that the equations of the paths of A and B are:

$$\text{For A: } y = -\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha$$

$$\text{For B: } y = -\frac{gx^2}{2V^2} \sec^2 \beta + x \tan \beta$$

Show that the x -coordinate of the point R of collision is

$$x = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}. \quad 3$$

(ii) You may assume that the equation of the horizontal displacement of A is $x = VT \cos \alpha$.

(α) Write down the equation for the horizontal displacement of B (Remember that B is projected T seconds after A). 1

(β) Show that the difference T in the times of projection is

$$T = \frac{2V(\cos \beta - \cos \alpha)}{g \sin(\alpha + \beta)}. \quad 2$$

Question 4:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) Use the table of standard integrals to find the area enclosed between the curve $y = \frac{1}{\sqrt{4-x^2}}$, the x -axis and the ordinates at $x = -1$ and $x = 1$. **2**
- (b) Find the volume of the solid of revolution formed when the area enclosed by the curve $y = \frac{1}{\sqrt{4-x^2}}$ and the x -axis between the ordinates at $x = -1$ and $x = 1$ is rotated around the x -axis. **3**
- (c) Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{2-2\cos 2x} \, dx$. **2**
- (d) Evaluate $\int_0^1 x\sqrt{1-x} \, dx$ **3**
- (e) (i) If $\mathbf{I}_n = \int_1^e x(\ln x)^n \, dx$; $n = 0, 1, 2, 3, \dots$. **5**
Show that $\mathbf{I}_n = \frac{e^2}{2} - \frac{n}{2} \mathbf{I}_{n-1}$; $n = 0, 1, 2, 3, \dots$
- (ii) Hence evaluate $\int_1^e x(\ln x)^3 \, dx$

Question 5:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) Two complex numbers are given by
$$z = 3 - 4i \text{ and } w = 2 - 2i.$$
6
- (i) Find the values of the product $\bar{z}w$.
- (ii) Find the two square roots of z .
- (iii) Express w in modulus-argument form and hence find the value of w^4 .
- (b) The locus of a point P which moves in the complex plane is represented by the equation $|z - (3 + 4i)| = 5$. **4**
- (i) Sketch the locus of the point P .
- (ii) Find the maximum value of the modulus of z and write down the value of $\arg z$ when P is in the position of maximum modulus.
- (iii) Find the modulus of z when $\arg z = \tan^{-1}\left(\frac{1}{2}\right)$.
- (c) Using the complex number $z = 1 - i$:
find the real numbers a and b such that $az = 1 - \frac{b}{z}$ **2**
- (d) In an Argand diagram, the point P represents the complex number z , the point Q represents the complex number w and the point R represents the complex number $z - w$ with O being the origin.
It is given that $z - w = iz$. **3**
- (i) Describe the geometric properties of the triangle POR , providing full reasoning for your answer.
- (ii) Find the size of the angle QPR .

Question 6: Use a SEPARATE Writing Booklet.

Total Marks 15

- (a) Consider the curve defined by $2x^2 + xy - y^2 = 0$. At the point $(2, 4)$ on the curve, find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. **4**

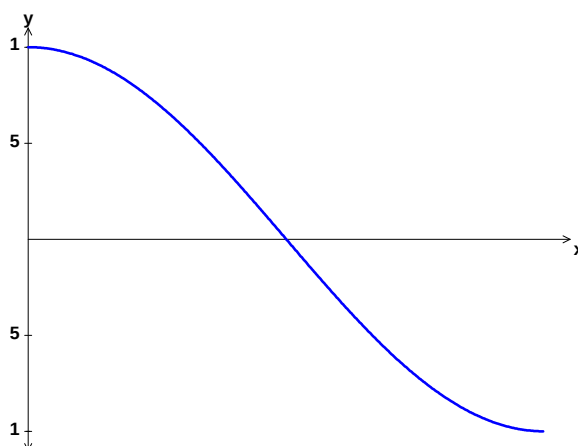
- (b) Consider the function $f(x) = \frac{x}{1-x^2}$.

(i) Show that the function is increasing for all values of x in its domain. **1**

(ii) Sketch the graph of $y = f(x)$ showing the intercepts on the axes and the equations of any asymptotes **2**

(iii) Find the values of k such that the equation $\frac{x}{1-x^2} = kx$ has three distinct real roots. **2**

(c)



The graph of $f(x) = \cos x$ is shown above in the interval $0 \leq x \leq \pi$.

Provide separate half-page sketches of the graphs of the following:

(i) $y = [f(x)]^2$ **2**

(ii) $y^2 = f(x)$ **2**

(iii) $y = \frac{1}{|f(x)|}$ **2**

Question 7:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) Solve the equation $|z|^2 + 2i\bar{z} = 4i + 7$, expressing any answers in the form $z = a + ib$ where a and b are real. **3**

- (b) A, B and C are the angles of a triangle.
Show that $(\cos A + i \sin A)(\cos B + i \sin B) + (\cos C - i \sin C) = 0$ **3**

- (c) Using $z = \cos \theta + i \sin \theta$

- (i) Express $1 + z$ in modulus-argument form.

Hence show that $(1 + z)^4 = 16 \cos^4\left(\frac{\theta}{2}\right)(\cos 2\theta + i \sin 2\theta)$. **3**

- (ii) Use the Binomial Theorem expansion of $(1 + z)^4$ to show that

$$1 + 4 \cos \theta + 6 \cos 2\theta + 4 \cos 3\theta + \cos 4\theta = 16 \cos^4\left(\frac{\theta}{2}\right) \cos 2\theta,$$

and find a corresponding expression for

$$4 \sin \theta + 6 \sin 2\theta + 4 \sin 3\theta + \sin 4\theta. \quad \text{3}$$

- (iii) Hence show that

$$\frac{4 \sin \theta + 6 \sin 2\theta + 4 \sin 3\theta + \sin 4\theta}{1 + 4 \cos \theta + 6 \cos 2\theta + 4 \cos 3\theta + \cos 4\theta} = \tan 2\theta,$$

and

$$\frac{4 \sin \theta + 4 \sin 3\theta + \sin 4\theta}{1 + 4 \cos \theta + 4 \cos 3\theta + \cos 4\theta} = \tan 2\theta. \quad \text{3}$$

Question 8:*Use a SEPARATE Writing Booklet.***Total Marks 15**

- (a) (i) Use the substitution $x = 10\sqrt{2} \sin \theta$ to show that

$$\int_{-10}^{10} \sqrt{200 - x^2} \, dx = 100 + 50\pi,$$

then use a geometrical argument to verify this result.

4

- (ii) A mould from a model railway tunnel is made by rotating the region bounded by the curve $y = \sqrt{200 - x^2}$ and the x -axis between the lines $x = -10$ and $x = 10$ through 180° about the line $x = 100$ (where all measurements are in cm). Use the method of cylindrical shells to show that the volume $V \text{ cm}^3$ of the tunnel is given by

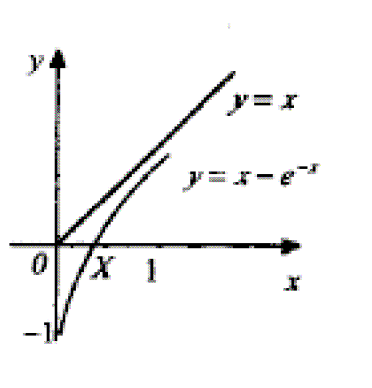
$$V = \pi \int_{-10}^{10} (100 - x) \sqrt{200 - x^2} \, dx.$$

Hence find the volume of the tunnel in m^3 correct to 2 significant figures.

4

Question 8 continued on next page

(b) (Continued from previous page)



The diagram shows the graph of the curve $y = x - e^{-x}$, $x \geq 0$.

This curve makes an intercept X on the x -axis, where $0 < X < 1$.

The region bounded by the curve and the line $y = x$

between $x = 0$ and $x = X$ is rotated through

one complete revolution about the y -axis.

- (i) Use the method of cylindrical shells to show that the volume V of the solid of revolution is given by

$$V = 2\pi \int_0^X x e^{-x} dx . \quad 3$$

- (ii) Hence show that $V = 2\pi(1 - X - X^2)$. 4

THE END

THIS PAGE IS BLANK

EXTENSION 2 MATHEMATICS - HALF-YEARLY 2008

SOLUTIONS - PR

Question 1:

$$\begin{aligned}
 \text{(a) (i)} \quad \frac{dv}{dt} &= \frac{dv}{dx} \times \frac{dx}{dt} \quad \checkmark \\
 &= \frac{dv}{dx} \cdot (v) \\
 &= \frac{dv}{dx} \cdot \frac{d(\frac{1}{2}v^2)}{dv} = \frac{d(\frac{1}{2}v^2)}{dx} \quad \checkmark
 \end{aligned}$$

$$\text{(ii)} \quad v^2 = 4x - 2x^2$$

$$\text{(a)} \quad v=0 \quad \therefore 0 = 2x(2-x) \quad \checkmark$$

$$\therefore x=0, x=2$$

$$\therefore \text{The amplitude is } \frac{0+2}{2} = 1 \quad \checkmark$$

$$\begin{aligned}
 \text{(b)} \quad v^2 &= 4x - 2x^2 \\
 &= 2(2x - x^2)
 \end{aligned}$$

$$\therefore \frac{1}{2}v^2 = 2x - x^2$$

$$a = \ddot{x} = 2 - 2x \quad (a = -n^2x) \quad \checkmark$$

$$= 2(1-x)$$

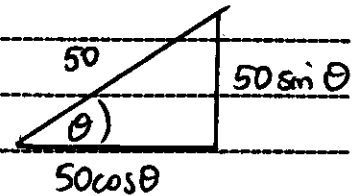
$$= -2(x-1)$$

$$\therefore n^2 = 2$$

$$\therefore n = \sqrt{2}$$

$$\text{Period: } T = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi \quad \checkmark$$

(b) (i) $\ddot{x} = 0$
 $\dot{x} = c$



when $t=0$, $\dot{x} = 50 \cos \theta \therefore c = 50 \cos \theta$

$\therefore \dot{x} = 50 \cos \theta$ ✓

$x = \int \dot{x} dt = \int 50 \cos \theta dt = 50t \cos \theta + c$

when $t=0$, $x=0 \therefore c=0$

$\therefore x = 50t \cos \theta$ ✓

$\ddot{y} = -10$

$\dot{y} = \int -10 dt = -10t + c$

when $t=0$, $\dot{y} = 50 \sin \theta \therefore c = 50 \sin \theta$

$\dot{y} = -10t + 50 \sin \theta$ ✓

$y = \int (-10t + 50 \sin \theta) dt = -5t^2 + 50t \sin \theta + c$

when $t=0$, $y=80 \therefore c=80$

$\therefore y = -5t^2 + 50t \sin \theta + 80$ ✓

(ii) when $x=300$, $y=2$

$\therefore 300 = 50t \cos \theta$

$t = 6/\cos \theta$ ✓

Sub into horizontal equation: $2 = -5 \left(\frac{6}{\cos \theta} \right)^2 + 50 \left(\frac{6}{\cos \theta} \right) \sin \theta + 80$ ✓

$\therefore 0 = -180 \sec^2 \theta + 300 \tan \theta + 78$

$0 = -180(\tan^2 \theta + 1) + 300 \tan \theta + 78$

$= -180 \tan^2 \theta + 300 \tan \theta - 102$

$0 = 180 \tan^2 \theta - 300 \tan \theta + 102$ ✓

$\therefore \tan \theta = \frac{300 \pm \sqrt{300^2 - 4 \times 180 \times 102}}{2 \times 180}$

$\therefore 1.19$ or 0.475

$\therefore \theta = 49^\circ 58' \text{ or } 25^\circ 27'$ ✓ ✓

$\therefore$ Initial angle of projection could be either 25 or 50 degrees.

Question 2:

$$(a) \quad (2 + \cos x)(2 - \cos y) = 3$$

$$2 - \cos y = \frac{3}{2 + \cos x}$$

$$\therefore \cos y = 2 - \frac{3}{2 + \cos x} \quad \checkmark$$

$$\cos y = \frac{4 + 2\cos x - 3}{2 + \cos x}$$

$$= \frac{1 + 2\cos x}{2 + \cos x}$$

$$\cos^2 y + \sin^2 y = 1$$

$$\therefore \sin^2 y = 1 - \left(\frac{1 + 2\cos x}{2 + \cos x} \right)^2$$

$$= \frac{(2 + \cos x)^2 - (1 + 2\cos x)^2}{(2 + \cos x)^2} \quad \checkmark$$

$$= \frac{3 - 3\cos^2 x}{(2 + \cos x)^2}$$

$$= \frac{3(1 - \cos^2 x)}{(2 + \cos x)^2}$$

$$\sin^2 y = \frac{3\sin^2 x}{(2 + \cos x)^2}$$

$$\therefore \sin y = \frac{\sqrt{3} \sin x}{2 + \cos x}$$

$$(ii) \quad (2 + \cos x)(2 - \cos y) = 3$$

Product rule using implicit differentiation.

$$u = 2 + \cos x \quad \therefore u' = -\sin x \quad \checkmark$$

$$v = 2 - \cos y \quad \therefore v' = \sin y \left(\frac{dy}{dx} \right)$$

$$\therefore -\sin x (2 - \cos y) + (2 + \cos x) \sin y \left(\frac{dy}{dx} \right) = 0$$

$$\therefore \frac{dy}{dx} = \frac{\sin x (2 - \cos y)}{\sin y (2 + \cos x)} \quad \checkmark$$

Substituting for $\sin y$: $\frac{\sin y}{\sin x} = \frac{\sqrt{3}}{2 + \cos x}$

$$\therefore \frac{dy}{dx} = \left(\frac{2 + \cos x}{\sqrt{3}} \right) \left(\frac{2 - \cos y}{2 + \cos y} \right)$$

but $(2 + \cos x)(2 - \cos y) = 3$ ✓

$$\therefore \frac{dy}{dx} = \frac{3}{\sqrt{3}(2 + \cos y)}$$

$$= \frac{\sqrt{3}}{2 + \cos y}$$

(b)(i) Even function if $f(-x) = f(x)$

$$\therefore f(-x) = a + \frac{b \sin(-x)}{-x}$$
 ✓

$$= a + \frac{-b \sin x}{-x}$$

$$= a + \frac{b \sin x}{x}$$

$$= f(x) \therefore f(x) \text{ is even}$$

(ii) $f(x) = a$

$$\therefore a = a + \frac{b \sin x}{x}$$

$$\therefore 0 = \frac{b \sin x}{x} \quad x \neq 0$$

$$\therefore \sin x = 0$$
 ✓

$$\therefore x = n\pi, \quad n = \pm 1, \pm 2, \pm 3, \dots$$
 ✓

(iii) $\left| \frac{b \sin x}{x} \right| \leq \left| \frac{b}{x} \right| \rightarrow \text{as } x \rightarrow \infty$

$$\therefore \lim_{x \rightarrow \infty} f(x) = a < a = 1$$
 ✓

$$\lim_{x \rightarrow 0} f(x) = a + b \lim_{x \rightarrow 0} \frac{\sin x}{x} = a + b \quad \therefore b = -1$$
 ✓

$$(c) \int \frac{1}{1-x^2} dx = \int \frac{1}{(1+x)(1-x)} dx$$

Partial fractions: $\frac{a}{(1+x)} + \frac{b}{(1-x)} = \frac{1}{(1+x)(1-x)}$

$$\therefore a(1-x) + b(1+x) = 1$$

$$\therefore a+b + x(b-a) = 1$$

$$\therefore a+b=1 \text{ and } b-a=0$$

$$\therefore a=b \text{ and } 2b=1 \quad b=\frac{1}{2}$$

$$\therefore \int \frac{1}{1-x^2} dx = \int \left(\frac{1/2}{1+x} + \frac{1/2}{1-x} \right) dx \quad \checkmark$$

$$= \frac{1}{2} \left(\ln|1+x| - \ln|1-x| \right) + c$$

$$= \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + c \quad \checkmark$$

$$(d) \int \frac{e^{2x}}{(e^x-1)^2} dx$$

let $u = e^x - 1 \Rightarrow e^x = u+1$
 $\frac{du}{dx} = e^x \Rightarrow du = e^x dx$

$$= \int \frac{e^x}{(e^x-1)^2} e^x dx$$

$$= \int \frac{u+1}{u^2} du \quad \checkmark$$

$$= \int \left(\frac{u}{u^2} + \frac{1}{u^2} \right) du$$

$$= \int \left(\frac{1}{u} + \frac{1}{u^2} \right) du$$

$$= \ln|u| - \frac{1}{u} + c \quad \checkmark$$

$$= \ln|e^x-1| - \frac{1}{e^x-1} + c \quad \checkmark$$

Question 3:

(a) (i) Gradient OP = $y_1/x_1 = \frac{(\ln x_1)^2}{x_1}$

$y = (\ln x)^2 \therefore \frac{dy}{dx} = \frac{2 \ln x}{x} @ x = x_1$

$\therefore \frac{dy}{dx} = \frac{2 \ln x_1}{x_1}$

Equate gradients $\therefore \frac{(\ln x_1)^2}{x_1} = \frac{2 \ln x_1}{x_1}$ ✓

$\therefore (\ln x_1)^2 - 2 \ln x_1 = 0$

$\ln x_1 (\ln x_1 - 2) = 0$ ✓

$\therefore \ln x_1 = 0$ (Not possible) $\approx \ln x_1 = 2$

$\ln x_1 = 2$

$\therefore x_1 = e^2 \quad y_1 = (\ln e^2)^2$
 $= 2^2$
 $= 4$

$\therefore P(e^2, 4)$

(ii) $f(x) = kx$ has two distinct real roots if the line $y = kx$ cuts the curve in two points

$\therefore 0 < k < \text{gradient OP}$

$0 < k < 4e^{-2}$ ✓

(iii) $\int (\ln x)^2 dx = \int 1 \times (\ln x)^2 dx$ $u = x \quad v = (\ln x)^2$

$u' = 1 \quad v' = \frac{2 \ln x}{x}$

$= x(\ln x)^2 - \int x \cdot \frac{2 \ln x}{x} dx$ ✓

$= x(\ln x)^2 - \int 2 \ln x dx$

$u = x \quad v = \ln x$

$= x(\ln x)^2 - 2 \left[x \ln x - \int x \cdot \frac{1}{x} dx \right]$ ✓ $u' = 1 \quad v' = 1/x$

$= x(\ln x)^2 - 2x \ln x + 2x + C$

$$A = \frac{1}{2} e^2 \cdot 4 - \int_1^{e^2} (\ln x)^2 dx$$

$$= 2e^2 - \left[x(\ln x)^2 - 2x \ln x + 2x \right]_1^{e^2} \quad \checkmark$$

$$= 2e^2 - (e^2 (\ln e^2)^2 - 2e^2 \ln e^2 + 2e^2 - (0 - 0 + 2))$$

$$= 2e^2 - (4e^2 - 4e^2 + 2e^2 - 2)$$

$$= 2 \quad \checkmark$$

Area is 2 square units

$$(iv) \quad y = (\ln x)^2 \quad x \geq 1$$

$$\sqrt{y} = \ln x$$

$$e^{\sqrt{y}} = x$$

$$\therefore f^{-1}(x) = e^{\sqrt{x}} \quad \checkmark$$

(v) The required tangent is the reflection of OP in the line $y=x$.
It passes through $(4, e^2)$ and has equation

$$y = \frac{1}{4} e^2 x \quad \checkmark$$

(b) (i) Equate equations for A and B

$$\frac{-gx^2 \sec^2 \alpha + x \tan \alpha}{2v^2} = \frac{-gx^2 \sec^2 \beta + x \tan \beta}{2v^2} \quad \checkmark$$

$$\therefore \frac{gx^2}{2v^2} (\sec^2 \alpha - \sec^2 \beta) = x (\tan \alpha - \tan \beta)$$

$$\frac{gx}{2v^2} (\tan^2 \alpha - \tan^2 \beta) = (\tan \alpha - \tan \beta) \quad x \neq 0 \quad \checkmark$$

$$\therefore \frac{gx}{2v^2} = \frac{\tan \alpha - \tan \beta}{\tan^2 \alpha - \tan^2 \beta}$$

$$\begin{aligned}
 \therefore x &= \frac{2V^2}{g} \times \left(\frac{1}{(\tan \alpha + \tan \beta)} \right) \quad \tan \alpha \neq \tan \beta \\
 &= \frac{2V^2}{g} \left(\frac{\cos \alpha \cos \beta}{\sin \alpha \cos \beta + \cos \alpha \sin \beta} \right) \\
 &= \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)} \quad \checkmark
 \end{aligned}$$

(ii) (α) $x = VT \cos \alpha$

New equation: $x = V(t - T) \cos \beta \quad \checkmark$

(β) when A is at R:

$$VT \cos \alpha = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}$$

$$\therefore t = \frac{2V \cos \beta}{g \sin(\alpha + \beta)} \quad \checkmark$$

When B is at R:

$$V(t - T) \cos \beta = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}$$

$$t - T = \frac{2V \cos \alpha}{g \sin(\alpha + \beta)}$$

$$\therefore T = t - \frac{2V \cos \alpha}{g \sin(\alpha + \beta)}$$

$$= \frac{2V \cos \beta}{g \sin(\alpha + \beta)} - \frac{2V \cos \alpha}{g \sin(\alpha + \beta)} \quad \checkmark$$

$$= \frac{2V(\cos \beta - \cos \alpha)}{g \sin(\alpha + \beta)}$$

Question 4:

$$(a) A = \int_{-1}^1 \frac{1}{\sqrt{4-x^2}} dx \quad \text{Even Function}$$

$$= 2 \int_0^1 \frac{1}{\sqrt{4-x^2}} dx$$

$$= 2 \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^1 \quad \checkmark$$

$$= 2 \left(\sin^{-1}(1/2) - \sin^{-1}(0) \right)$$

$$= 2 \left(\pi/6 \right)$$

$$= \pi/3 \quad \text{units}^2 \quad \checkmark$$

$$(b) V = \pi \int_{-1}^1 \left(\frac{1}{\sqrt{4-x^2}} \right)^2 dx \quad \checkmark$$

$$= \pi \int_{-1}^1 \frac{1}{4-x^2} dx$$

$$= \pi \int_{-1}^1 \left(\frac{a}{2-x} \right) + \left(\frac{b}{2+x} \right) dx$$

$$a(2+x) + b(2-x) \equiv 1$$

$$2a - bx + 2a + 2b \equiv 1$$

$$= \frac{\pi}{4} \int_{-1}^1 \left(\frac{1}{2-x} + \frac{1}{2+x} \right) dx \quad \checkmark$$

$$\therefore a - b = 0 \quad \text{and} \quad 2a + 2b = 1$$

$$\therefore a = b \quad a + b = 1/2$$

$$\therefore a = b = 1/4$$

$$= \frac{\pi}{4} \left[\ln \left| \frac{2+x}{2-x} \right| \right]_{-1}^1$$

$$= \frac{\pi}{4} \left(\ln 3 - \ln(1/3) \right)$$

$$= \frac{2 \ln 3 \pi}{4}$$

$$= \frac{\pi \ln 3}{2} \quad \text{units}^3 (= 1.73) \quad \checkmark$$

$$\begin{aligned}
 (c) \quad \int_0^{\pi/2} \sqrt{2-2\cos 2x} \, dx &= \int_0^{\pi/2} \sqrt{2} \sqrt{1-\cos 2x} \, dx \\
 &= \int_0^{\pi/2} \sqrt{2} \sqrt{2\sin^2 x} \, dx \quad \checkmark \\
 &= \sqrt{2} \int_0^{\pi/2} \sqrt{2} \sin x \, dx \\
 &= 2 \int_0^{\pi/2} \sin x \, dx \\
 &= 2 \left[-\cos x \right]_0^{\pi/2} dx \\
 &= 2 (0 - -1) \\
 &= 2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad \int_0^1 x\sqrt{1-x} \, dx & \quad \text{let } u=1-x \quad \therefore x=1-u \\
 & \quad \begin{array}{lll} x=1 & u=0 & du=-1 \\ x=0 & u=1 & dx \end{array} \\
 & \quad \frac{du}{dx} = -1 \\
 & \quad dx = -du \\
 &= \int_1^0 (1-u) u^{1/2} \cdot -du \quad \checkmark \\
 &= - \int_1^0 (u^{1/2} - u^{3/2}) du \\
 &= \int_0^1 (u^{1/2} - u^{3/2}) du \\
 &= \left[\frac{2u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right]_0^1 \quad \checkmark \\
 &= \left(\frac{2}{3} - \frac{2}{5} \right) - (0) \\
 &= \frac{4}{15} \quad \checkmark
 \end{aligned}$$

(e) (i) $I_n = \int_1^e x (\ln x)^n dx \quad n=0,1,2,3 \dots$

$$= \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \int_1^e \frac{x^2}{2} \cdot \frac{n}{x} (\ln x)^{n-1} dx \quad u = \frac{x^2}{2} \quad v = (\ln x)^n$$

$$u' = x \quad v' = \frac{n}{x} (\ln x)^{n-1}$$

$$= \left(\frac{e^2}{2} - 0 \right) - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

(ii) $\int_1^e x (\ln x)^3 dx$ implies I_3

$$I_3 = \frac{e^2}{2} - \frac{3}{2} I_2$$

$$= \frac{e^2}{2} - \frac{3}{2} \left[\frac{e^2}{2} - \frac{2}{2} I_1 \right]$$

$$= \frac{e^2}{2} - \frac{3}{2} \left[\frac{e^2}{2} - \left(\frac{e^2}{2} - \frac{1}{2} I_0 \right) \right]$$

$$= \frac{e^2}{2} - \frac{3}{4} I_0$$

$$= \frac{e^2}{2} - \frac{3}{4} \left(\frac{e^2}{2} - \frac{1}{2} \right) \quad I_0 = \int_1^e x (\ln x)^0 dx$$

$$= \frac{e^2}{2} - \frac{3e^2}{8} + \frac{3}{8} \quad = \int_1^e x dx$$

$$= \frac{e^2}{8} + \frac{3}{8} \quad = \frac{e^2}{2} - \frac{1}{2}$$

Question 5:

$$\begin{aligned}
 (a) (i) \bar{z}w &= (3+4i)(2-2i) \\
 &= 6+8i-6i-8i^2 \\
 &= 14+2i \quad \checkmark
 \end{aligned}$$

$$(ii) (a+ib)^2 = 3-4i \quad a, b \in \mathbb{R}$$

$$a^2 - b^2 + 2abi = 3 - 4i$$

$$\therefore a^2 - b^2 = 3 \quad \text{and} \quad ab = -2 \quad \checkmark$$

$$\therefore a = -2/b$$

$$\therefore (-2/b)^2 - b^2 = 3$$

$$4 - b^4 - 3b^2 = 0$$

$$b^4 + 3b^2 - 4 = 0$$

$$(b^2+4)(b^2-1) = 0$$

$$\therefore b = \pm 1 \quad b = \sqrt{-2} \text{ (Not applicable)}$$

$$b = +1 \quad a = -2$$

$$b = -1 \quad a = 2$$

$$\therefore \text{Roots are } -2+i \text{ and } 2-i \quad \checkmark$$

$$(iii) w = 2-2i$$

$$\begin{aligned}
 |w| &= \sqrt{4+4} \\
 &= 2\sqrt{2}
 \end{aligned}$$

$$\tan^{-1}\left(\frac{-2}{2}\right) = -\frac{\pi}{4}$$

$$\therefore w = 2\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \quad \checkmark$$

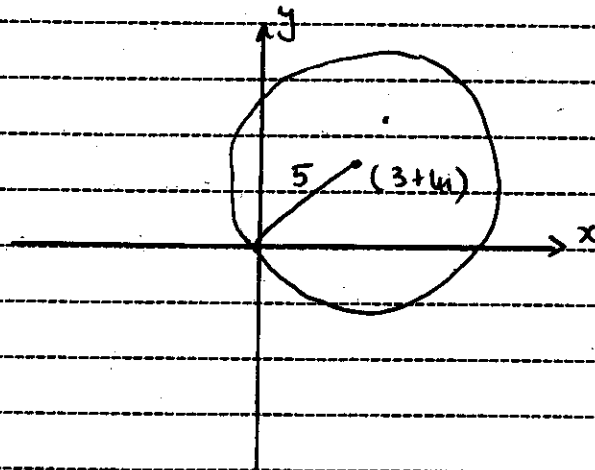
$$\therefore w^4 = (2\sqrt{2})^4 \operatorname{cis}\left(4\left(-\frac{\pi}{4}\right)\right) \quad \checkmark$$

$$= 64 \operatorname{cis}(-\pi)$$

$$= 64 \operatorname{cis} \pi$$

$$= -64 \quad \checkmark$$

(b) (i) $|z - (3+4i)| = 5$



(ii) maximum value of $|z| = 10$; diameter of the circle. $\arg z = \tan^{-1}(\frac{8}{6})$ ✓
 $z = 6+8i$ $= \tan^{-1}(\frac{4}{3})$

(iii) if $\arg z = \tan^{-1}(\frac{1}{2})$

$\therefore y = \frac{1}{2}x$ is a line

Need to find when this line intersects the circle

$$(x-3)^2 + (y-4)^2 = 25$$

sub for $y = \frac{x}{2}$

$$(x-3)^2 + (\frac{x}{2}-4)^2 = 25$$

$$x^2 - 6x + 9 + \frac{x^2}{4} - 4x + 16 = 25$$

$$\frac{5x^2}{4} - 10x = 0$$

$$x(\frac{5x}{4} - 10) = 0$$

$$\therefore x=0 \quad \text{or} \quad x=8$$

$$y=0$$

$$y=4$$

$\therefore$ intersect at $(0,0)$ and $(8,4)$

$$\therefore z = 8+4i$$

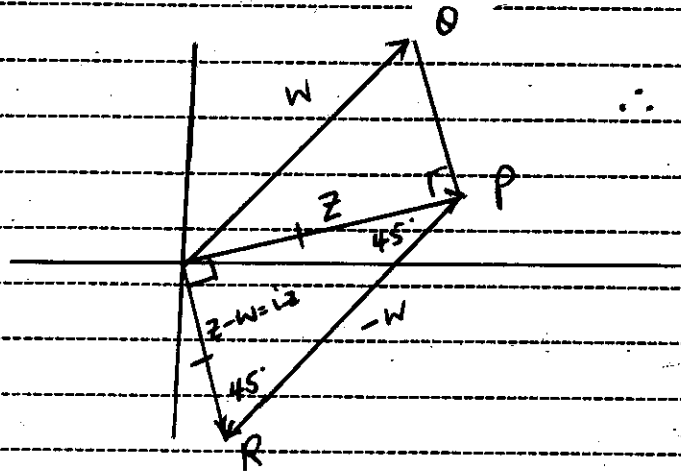
$$|z| = \sqrt{80}$$

$$= 4\sqrt{5}$$

(c) (i) $\triangle POR$ is an isosceles right angled triangle. ✓

$\begin{cases} |z| = |iz| \therefore OP = OR \therefore \text{isosceles} \\ OP \text{ is rotated through } 90^\circ \text{ anticlockwise to get } OR \end{cases}$ ✓

(ii)



$\therefore \angle PR = 135^\circ$ ✓
 $(= \frac{3\pi}{4})$

Question 6:

(a) $2x^2 + xy - y^2 = 0$

$$\therefore 4x + x \frac{dy}{dx} + y - 2y \frac{dy}{dx} = 0$$

$$4x + y = \frac{dy}{dx} (2y - x)$$

$$\therefore \frac{dy}{dx} = \frac{(4x + y)}{(2y - x)}$$

$$\frac{dy}{dx} \text{ @ } (2, 4) \quad \therefore \frac{dy}{dx} = \frac{4(2) + 4}{2(4) - 2} = \frac{12}{6} = 2 \quad \checkmark$$

$$\frac{d^2y}{dx^2} = \frac{(2y - x)(4 + \frac{dy}{dx}) - (4x + y)(2\frac{dy}{dx} - 1)}{(2y - x)^2} \quad \checkmark$$

at (2, 4)

$$\frac{d^2y}{dx^2} = \frac{6(6) - 12(3)}{36} = 0 \quad \checkmark$$

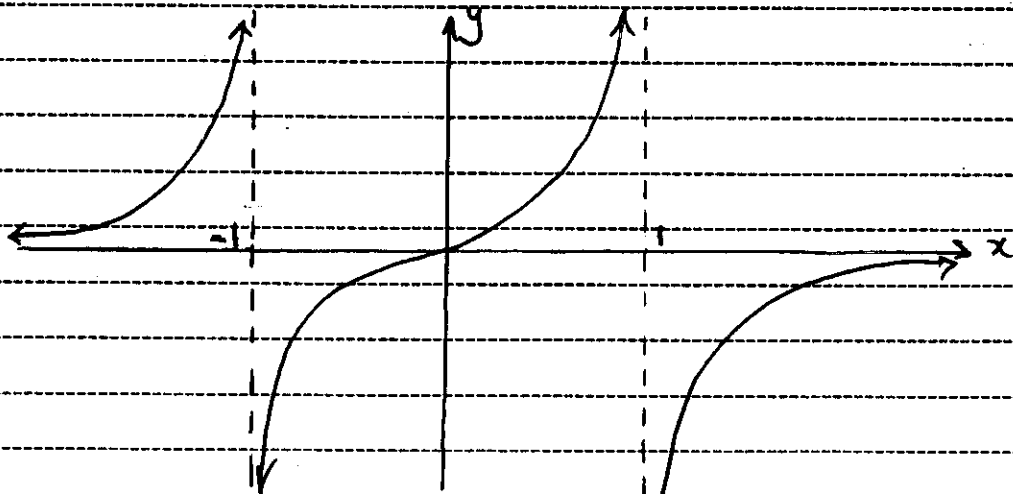
(b) (i) $f(x) = \frac{x}{1-x^2}$

$$f'(x) = \frac{(1-x^2)(1) - (x)(-2x)}{(1-x^2)^2}$$

$$f'(x) = \frac{x^2 + 1}{(1-x^2)^2} \quad \checkmark$$

 $f'(x)$ is always positive, so function always increasing.

(ii)



x = -1 Page No: _____

x = 1 ✓ (must have equations)

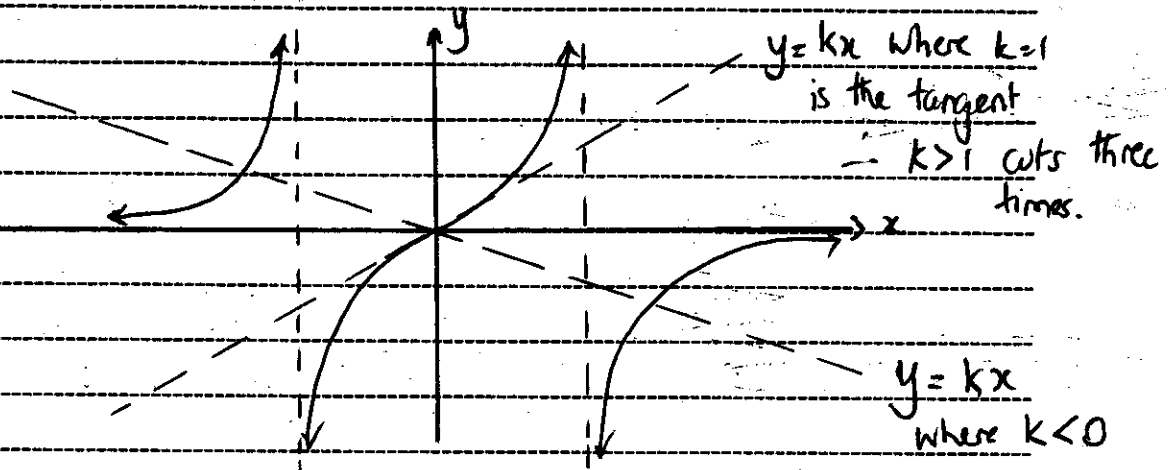
(ii) $\frac{x}{1-x^2} = kx$ has 3 distinct roots

$\therefore$ line $y=kx$ intersects $y=\frac{x}{1-x^2}$

three times $\Rightarrow$ 3 distinct roots

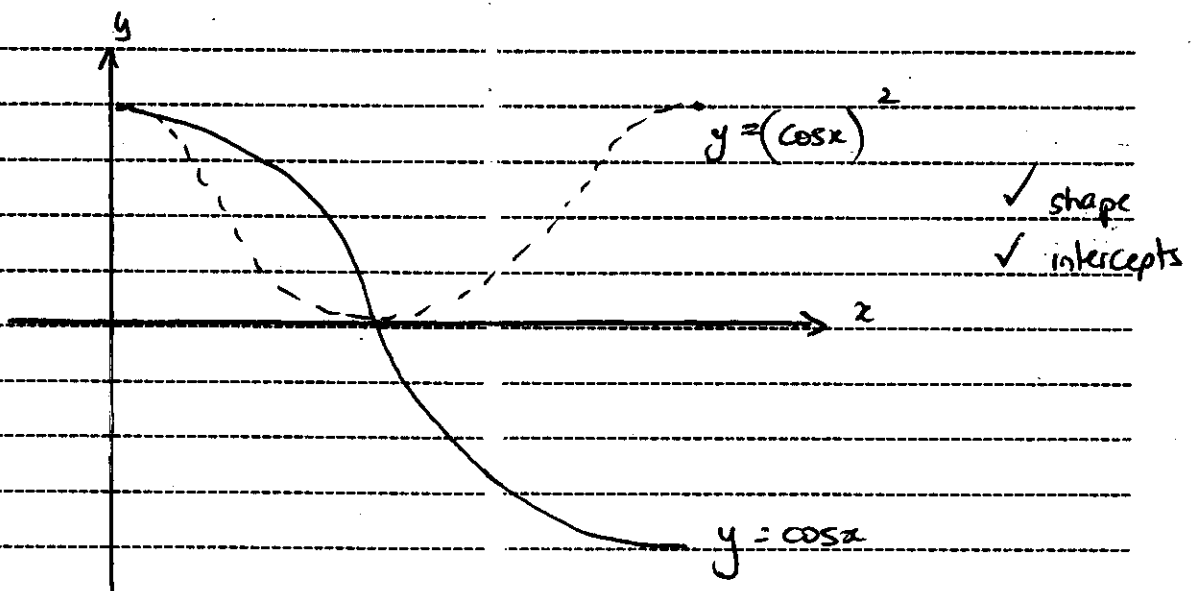
From the sketch below we can see:

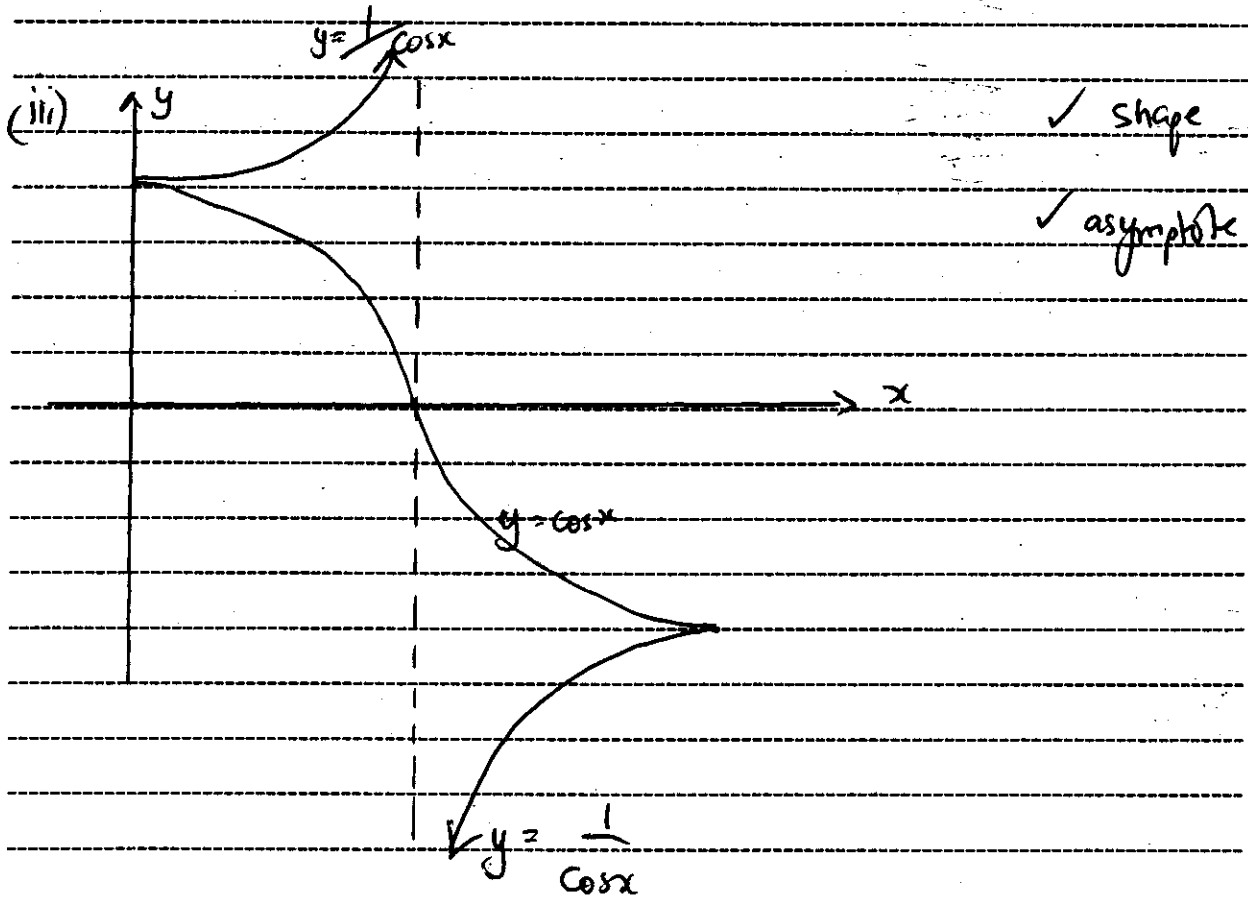
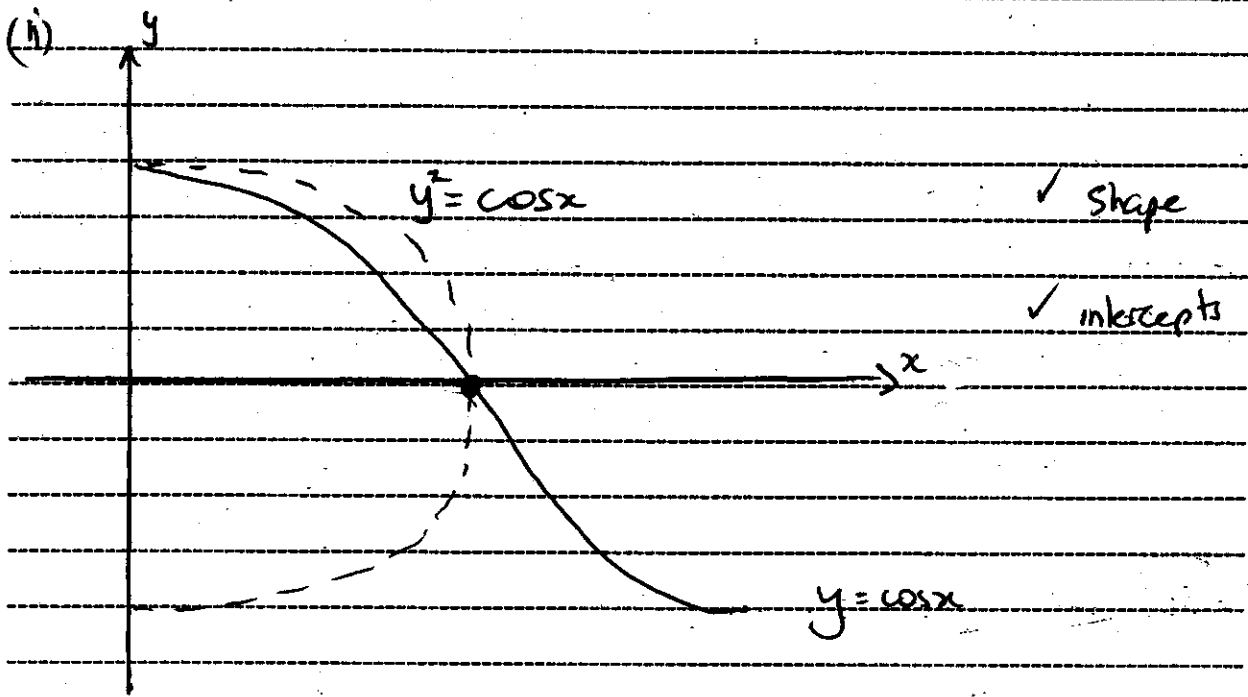
Either $k < 0$ or $k > 1$ as 3 roots ✓✓



(c)

(i)





Question 7:

$$(a) \quad |z|^2 + 2i\bar{z} = 4i + 7 \quad z = a + ib$$

$$a^2 + b^2 + 2i(a - ib) = 4i + 7 \quad \checkmark$$

$$2ai + a^2 + b^2 + 2b = 4i + 7$$

Equate real and imaginary parts

$$\therefore 2a = 4$$

$$a^2 + b^2 + 2b = 7 \quad \checkmark$$

$$a = 2$$

$$4 + b^2 + 2b = 7$$

$$b^2 + 2b - 3 = 0$$

$$(b + 3)(b - 1) = 0$$

$$b = -3 \quad b = 1$$

$\therefore$ The two solutions are: $z = 2 + i$ or $z = 2 - 3i \quad \checkmark$

$$(b) \quad (\cos A + i \sin A)(\cos B + i \sin B) + (\cos C - i \sin C) = 0$$

$$(\cos A + i \sin A)(\cos B + i \sin B) = \cos A \cos B - \sin A \sin B \quad \checkmark$$

$$+ i(\sin A \cos B + \cos A \sin B)$$

$$= \cos(A+B) + i \sin(A+B) \quad \checkmark$$

Now: $A+B = \pi - C$ [angle sum of triangle]

$$\therefore \cos(A+B) = \cos(\pi - C)$$

$$\cos(A+B) = -\cos C \quad \checkmark$$

and $\sin(A+B) = \sin(\pi - C)$

$$\sin(A+B) = \sin C$$

$$= -\cos C + i \sin C$$

$$= -(\cos C - i \sin C)$$

$$\therefore (\cos A + i \sin A)(\cos B + i \sin B) + (\cos C - i \sin C)$$

$$= -(\cos C - i \sin C) + (\cos C - i \sin C) = 0$$

$$\begin{aligned}
 \text{(c) (i)} \quad 1+z &= 1 + \cos\theta + i\sin\theta \\
 &= 2\cos^2\left(\frac{\theta}{2}\right) + i\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) \quad \checkmark \\
 &= 2\cos\left(\frac{\theta}{2}\right) \left[\cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right) \right] \quad \checkmark
 \end{aligned}$$

Use De Moivre's Theorem:

$$\begin{aligned}
 (1+z)^4 &= 2^4 \cos^4\left(\frac{\theta}{2}\right) \left(\cos\left(4\left(\frac{\theta}{2}\right)\right) + i\sin\left(4\left(\frac{\theta}{2}\right)\right) \right) \quad \checkmark \\
 &= 16 \cos^4\left(\frac{\theta}{2}\right) (\cos 2\theta + i\sin 2\theta)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad (1+z)^4 &= 1 + 4z + {}^4C_2 z^2 + {}^4C_3 z^3 + z^4 \\
 &= 1 + 4z + 6z^2 + 4z^3 + z^4
 \end{aligned}$$

$$\begin{aligned}
 z &= \cos\theta + i\sin\theta & z^2 &= \cos 2\theta + i\sin 2\theta & z^3 &= \cos 3\theta + i\sin 3\theta \\
 z^4 &= \cos 4\theta + i\sin 4\theta
 \end{aligned}$$

$$\begin{aligned}
 \therefore (1+z)^4 &= 1 + 4(\cos\theta + i\sin\theta) + 6(\cos 2\theta + i\sin 2\theta) \\
 &\quad + 4(\cos 3\theta + i\sin 3\theta) + (\cos 4\theta + i\sin 4\theta)
 \end{aligned}$$

$\therefore$ Equate real and imaginary components

$$\therefore 16 \cos^4\left(\frac{\theta}{2}\right) \cos 2\theta = 1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta \quad \checkmark$$

$$\therefore 16 \cos^4\left(\frac{\theta}{2}\right) \sin 2\theta = 4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta \quad \checkmark$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta}{1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta} &= \frac{16 \cos^4\left(\frac{\theta}{2}\right) \sin 2\theta}{16 \cos^4\left(\frac{\theta}{2}\right) \cos 2\theta} = \tan 2\theta \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \frac{4\sin\theta + 4\sin 3\theta + \sin 4\theta}{1 + 4\cos\theta + 4\cos 3\theta + \cos 4\theta} &= \frac{[16 \cos^4\left(\frac{\theta}{2}\right) - 6] \sin 2\theta}{[16 \cos^4\left(\frac{\theta}{2}\right) - 6] \cos 2\theta} = \tan 2\theta \quad \checkmark
 \end{aligned}$$

Question 8

$$(a) \quad (i) \quad x = 10\sqrt{2} \sin \theta \quad x = 10 \quad \theta = \pi/4$$

$$\therefore \frac{dx}{d\theta} = 10\sqrt{2} \cos \theta \quad x = -10 \quad \theta = -\pi/4$$

$$\therefore \int_{-10}^{10} \sqrt{200 - x^2} dx = \int_{-\pi/4}^{\pi/4} \sqrt{200 - 200 \sin^2 \theta} \cdot 10\sqrt{2} \cos \theta d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \sqrt{200} \cos \theta \cdot 10\sqrt{2} \cos \theta d\theta$$

$$= \int_{-\pi/4}^{\pi/4} 200 \cos^2 \theta d\theta$$

$$= 200 \int_{-\pi/4}^{\pi/4} \cos^2 \theta d\theta \quad \checkmark$$

$$= 100 \int_{-\pi/4}^{\pi/4} (1 + \cos 2\theta) d\theta$$

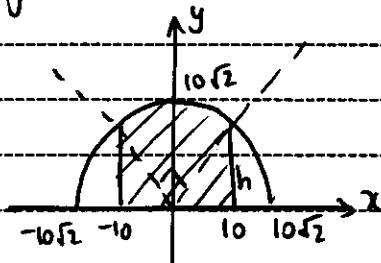
$$= 100 \left[\theta + \frac{1}{2} \sin 2\theta \right]_{-\pi/4}^{\pi/4} \quad \checkmark$$

$$= 100 \left[\left(\frac{\pi}{4} + \frac{1}{2} (1) \right) - \left(-\frac{\pi}{4} - \frac{1}{2} \right) \right]$$

$$= 100 \left(\frac{\pi}{2} + 1 \right)$$

$$= 100 + 50\pi \quad \checkmark$$

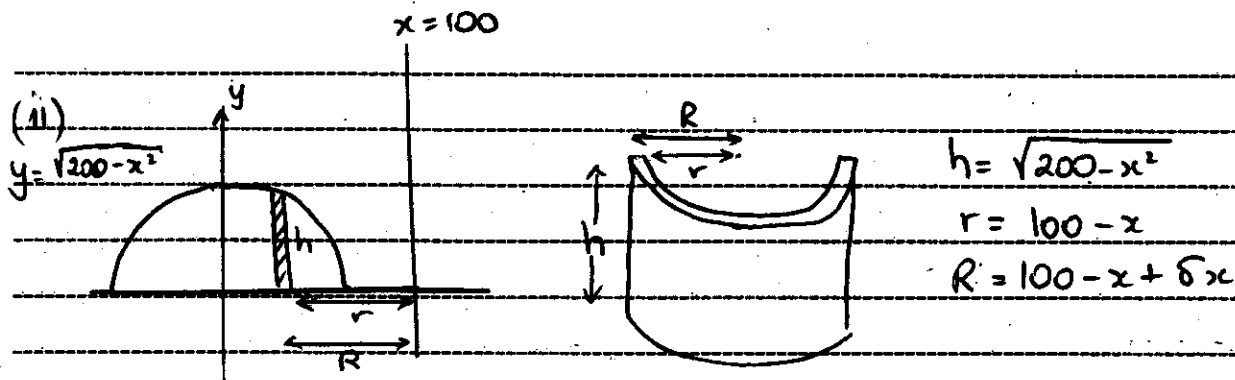
Diagram:



Geometrically integral gives shaded area comprising $\frac{1}{4}$ circle ($r = 10\sqrt{2}$) and two right triangles ($h = 10$). $\checkmark$

$$\text{Shaded area} = \frac{1}{4} \pi (10\sqrt{2})^2 + 2 \times \frac{1}{2} \times 10 \times 10$$

$$= 50\pi + 100$$



$$\delta V = \frac{1}{2} \pi (R^2 - r^2) h$$

$$= \frac{1}{2} \pi (R + r)(R - r) h$$

$$R - r = \delta x$$

$$= \frac{1}{2} \pi (200 - 2x + \delta x) \delta x h$$

$$= \frac{1}{2} \pi (2(100 - x) + \delta x) \delta x h$$

Ignoring terms in $(\delta x)^2$

$$\delta V = \pi (100 - x) h \delta x$$

$$= \pi (100 - x) \sqrt{200 - x^2} \delta x$$

$$\therefore V = \pi \lim_{\delta x \rightarrow 0} \sum_{-10}^{10} (100 - x) \sqrt{200 - x^2} \delta x$$

$$= \pi \int_{-10}^{10} (100 - x) \sqrt{200 - x^2} dx$$

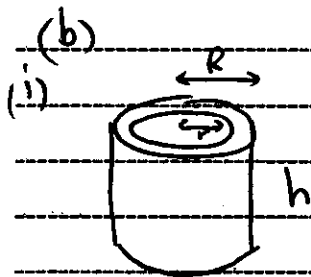
$$V = 100\pi \int_{-10}^{10} \sqrt{200 - x^2} dx - \pi \int_{-10}^{10} x \sqrt{200 - x^2} dx$$

$x \sqrt{200 - x^2}$ is an odd function, therefore integral is zero

$$\therefore V = 100\pi [100 + 50\pi] + 0$$

$$= 5000\pi(2 + \pi) \times 10^{-6} \text{ m}^3$$

$$= 0.081 \text{ m}^3$$



$$\begin{aligned}\delta V &= \pi (R^2 - r^2) h \\ &= \pi (R - r)(R + r) h \\ &= \pi (x + \delta x - x)(x + \delta x + x) e^{-x} \\ &= \pi (\delta x)(2x + \delta x) e^{-x} \quad \checkmark\end{aligned}$$

Ignoring second order terms

$$R = x + \delta x$$

$$r = x$$

$$\begin{aligned}h &= x - (x - e^{-x}) \\ &= e^{-x} \quad \checkmark\end{aligned}$$

$$\therefore \delta V = 2\pi e^{-x} x \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^x 2\pi x e^{-x} \delta x$$

$$V = 2\pi \int_0^x x e^{-x} dx \quad \checkmark$$

$$(ii) \quad \int_0^x x e^{-x} dx = \left[-x e^{-x} \right]_0^x + \int_0^x e^{-x} dx \quad \checkmark$$

$$= -x e^{-x} - \left[e^{-x} \right]_0^x \quad \checkmark$$

$$= -x e^{-x} - (e^{-x} - 1)$$

$$= 1 - e^{-x} - x e^{-x} \quad \checkmark$$

$$\text{But } 0 = x - e^{-x} \Rightarrow e^{-x} = x \quad \checkmark$$

$$\therefore V = 2\pi (1 - x - x^2)$$